Introduction to Prediction-based Inference

Methods & Applications

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Roadmap for today

Activity	**Time**
Prediction-based Inference (**PBI**) Overview	30 m
Break	5 m
Unit 00: Getting Started with `ipd`	30 m
Unit 01: AlphaFold Example	30 m
Wrap-Up and Q&A	10 m

Goals for this tutorial

1. Understand the fundamentals of PBI
2. Gain familiarity with PBI methods
3. Learn to run an analysis with predicted data using `ipd`

Goals for this tutorial

1. Understand the fundamentals of PBI

   → What is PBI?

   → Why is PBI important?

   → When did PBI become popular?

2. Gain familiarity with PBI methods

3. Learn to run an analysis with predicted data using `ipd`

What is PBI?

Statistical inference that is conducted with **predicted data**

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Statistical inference that is conducted with **predicted data**

- **Predicted data**: Data that is generated from a machine learning (ML) or Artificial Intelligence (AI) model rather than collected using the gold-standard method

Why is PBI important?

Gold-standard data is often infeasible or difficult to collect, while predicted data can be obtained quickly and inexpensively

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Gold-standard data is often infeasible or difficult to collect, while predicted data can be obtained quickly and inexpensively

$\downarrow$

Using predicted data can improve the **pace, cost, and scale** of scientific research

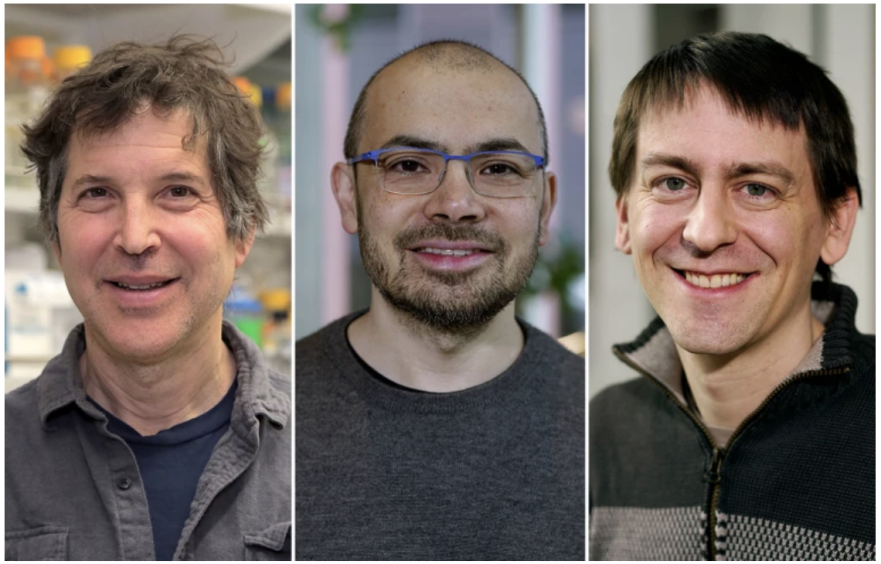
A highly influential example: AlphaFold

NEWS | 09 October 2024

Chemistry Nobel goes to developers of AlphaFold AI that predicts protein structures

This year's prize celebrates computational tools that have transformed biology and have the potential to revolutionize drug discovery.

By Ewen Callaway



David Baker, Demis Hassabis and John Jumper (left to right) won the chemistry Nobel for developing computational tools that can predict and design protein structures. Credit: BBVA Foundation

AlphaFold predictions are used for scientific inquiry

Google DeepMind

Targeting early-onset Parkinson's with AI

21 SEPTEMBER 2022

AlphaFold predictions are paving the way towards new treatments that can impact over 10 million people worldwide

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We'll use AlphaFold data later in the tutorial!

Using predicted data for EHR-based research

J Biomed Inform. 2024 Sep:157:104690. doi: 10.1016/j.jbi.2024.104690. Epub 2024 Jul 14.

Leveraging error-prone algorithm-derived phenotypes: Enhancing association studies for risk factors in EHR data

Yiwen Lu [1], Jiayi Tong [2], Jessica Chubak [3], Thomas Lumley [4], Rebecca A Hubbard [5], Hua Xu [6], Yong Chen [7]

Using predicted data for GWAS

Technical Report | Published: 13 June 2024

Synthetic surrogates improve power for genome-wide association studies of partially missing phenotypes in population biobanks

Zachary R. McCaw, Jianhui Gao, Xihong Lin & Jessica Gronsbell

Nature Genetics **56**, 1527–1536 (2024) | Cite this article

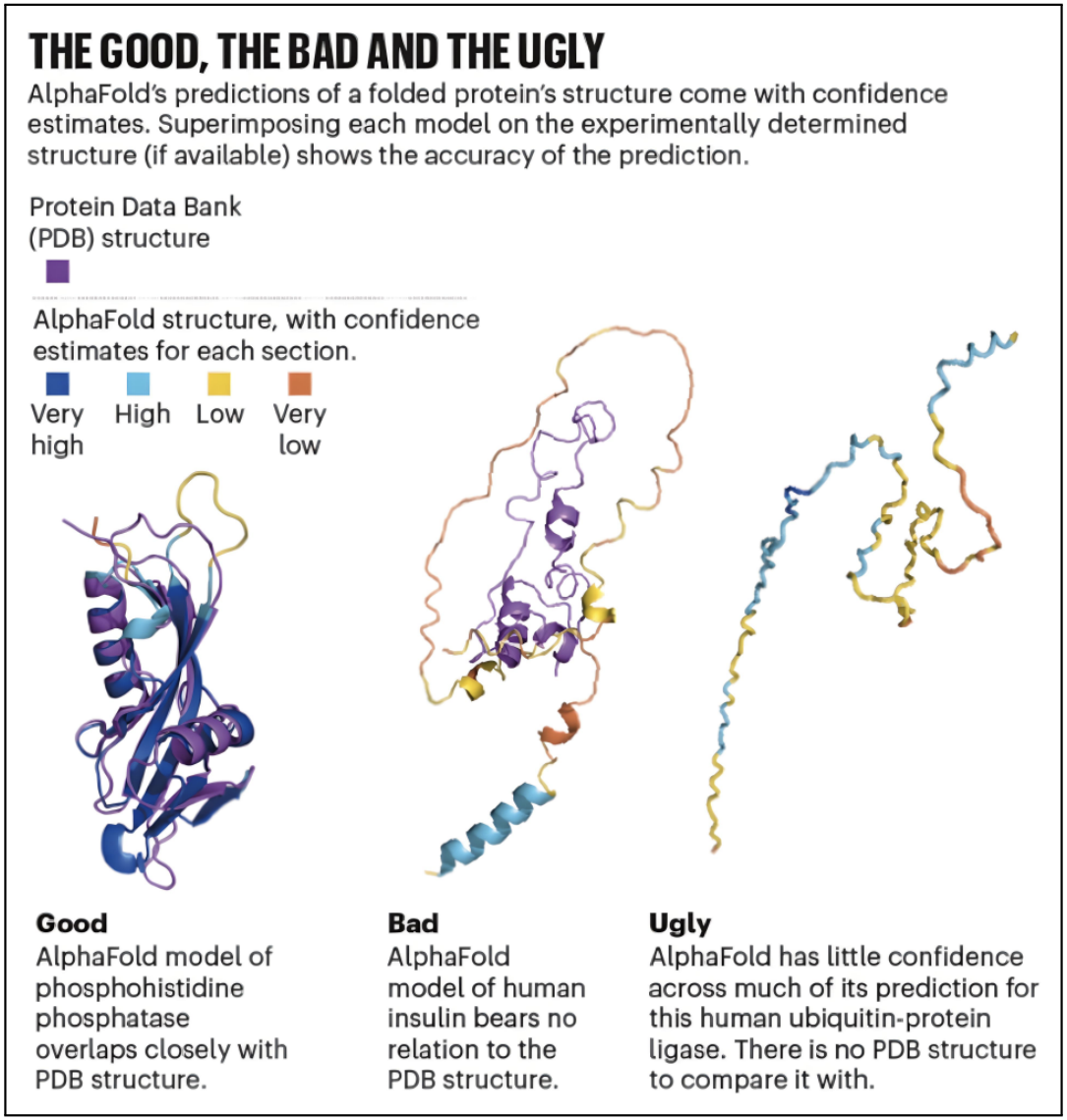
Using predicted data to study causes of death

From Narratives to Numbers:
Valid Inference Using Language Model Predictions From Verbal Autopsy Narratives

Shuxian Fan *[1] **Adam Visokay** *[1] **Kentaro Hoffman**[1] **Stephen Salerno**[2]
Li Liu[3] **Jeffrey T. Leek**[2] **Tyler H. McCormick**[1]

[1] University of Washington [2] Fred Hutchinson Cancer Center
[3]Johns Hopkins University

The catch: Predictions are imperfect



Thornton et al. Nature 2021.

More data, more danger

Scientific inquiry based on predicted data can lead us to be **more confident** in the **wrong answer**

When did PBI become popular?

Post-prediction inference (`postpi`)

RESEARCH ARTICLE | PHYSICAL SCIENCES |

Methods for correcting inference based on outcomes predicted by machine learning

Siruo Wang, Tyler H. McCormick, and Jeffrey T. Leek Authors Info & Affiliations

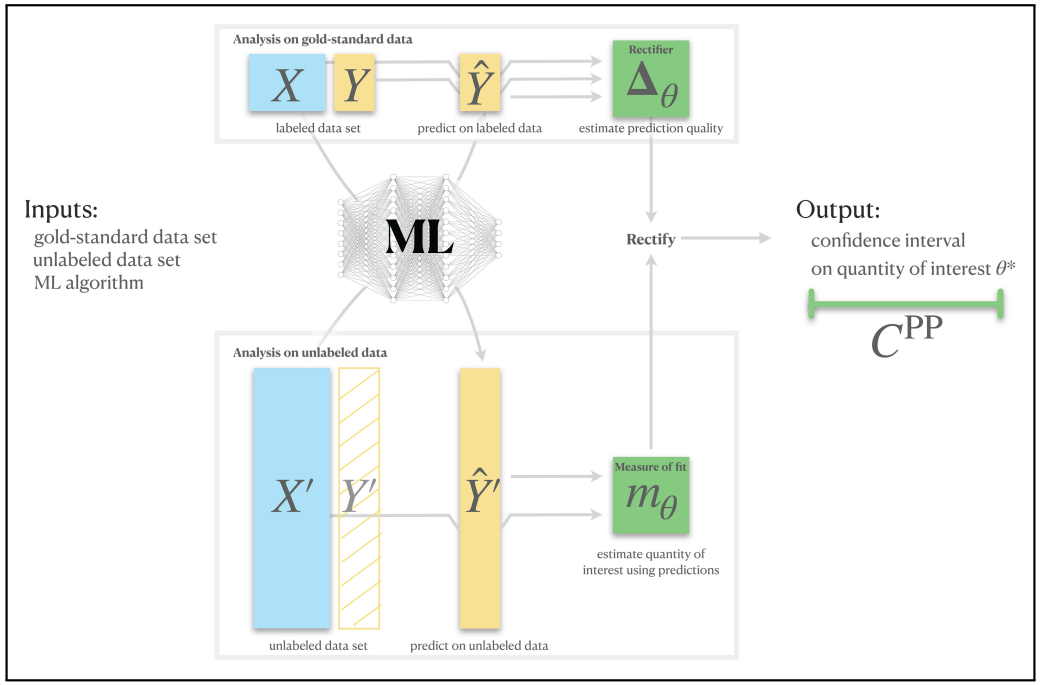
Edited by Robert Tibshirani, Stanford University, Stanford, CA, and approved October 6, 2020 (received for review January 24, 2020)

November 18, 2020 | 117 (48) 30266-30275 | https://doi.org/10.1073/pnas.2001238117

Goal: Mitigate bias introduced by using predictions as data

When did PBI become popular?

Prediction-powered inference (PPI)



Same goal, but a different approach

Angelopoulos, Bates, Fannjiang, Jordan, Zrnic. Prediction-powered inference. Science 2023.

Goals for this tutorial

1. Understand the fundamentals of PBI

2. Gain familiarity with PBI methods

   → What problem has been the focus of PBI?

   → What PBI methods currently exist?

3. Learn to run an analysis with predicted data using `ipd`

What problem has been the focus of PBI?

Regression

Understand the relationship between an outcome Y and covariates $\mathbf{X}$

The bad

Y is difficult to measure and observed in a subset of the data

What problem has been the focus of PBI?

Regression

Understand the relationship between an outcome Y and covariates $\mathbf{X}$

The bad

Y is difficult to measure and observed in a subset of the data

The good

- $\mathbf{X}$ is easy to measure and observed in the entire dataset
- A pre-trained ML model can produce predictions $\widehat{Y}$ of Y on the entire dataset

Necessary features of the predictions

1. **Derived from a pre-trained model**: The model was trained on a dataset that is independent of the dataset used for inference

Necessary features of the predictions

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   Examples

   - Openly available models (e.g., AlphaFold, GPT)
   - Models that you train (e.g., XGBoost, CNN)

Necessary features of the predictions

1. **Derived from a pre-trained model**: The model was trained on a dataset that is independent of the dataset used for inference

2. **Fully available**: The data $\mathbf{Z}$ used to train the model must be available for the entire dataset so may or may not include $\mathbf{X}$

Necessary features of the predictions

1. **Derived from a pre-trained model**: The model was trained on a dataset that is independent of the dataset used for inference

2. **Fully available**: The data $\mathbf{Z}$ used to train the model must be available for the entire dataset so may or may not include $\mathbf{X}$

   Chest X-ray Example: $\mathbf{Z}$ does not include $\mathbf{X}$
   - Y: True disease status
   - $\mathbf{X}$: Patient covariates (age, smoking, BMI)
   - $\mathbf{Z}$: Chest X-ray image
   - $\hat{Y}$: CNN prediction from the image

Necessary features of the predictions

1. **Derived from a pre-trained model**: The model was trained on a dataset that is independent of the dataset used for inference

2. **Fully available**: The data $\mathbf{Z}$ used to train the model must be available for the entire dataset so may or may not include $\mathbf{X}$

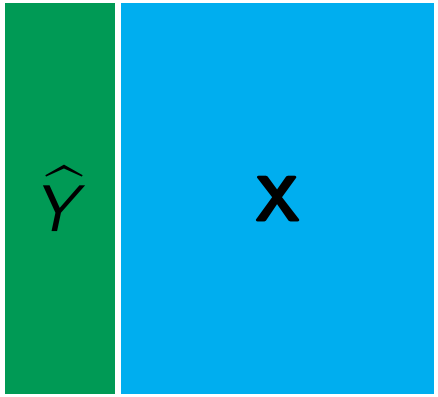
   GWAS Example: $\mathbf{Z}$ includes $\mathbf{X}$

   - Y: Body mass from DEXA
   - $\mathbf{X}$: SNPS, Genetic PCs, Age, Sex
   - $\mathbf{Z}$: Impedance measurements, Age, Sex
   - $\widehat{Y}$: Prediction from a random forest

A visual summary of the data

Labeled data

Unlabeled data



Y: Outcome
$\widehat{Y}$: Prediction
$\mathbf{X}$: Covariates

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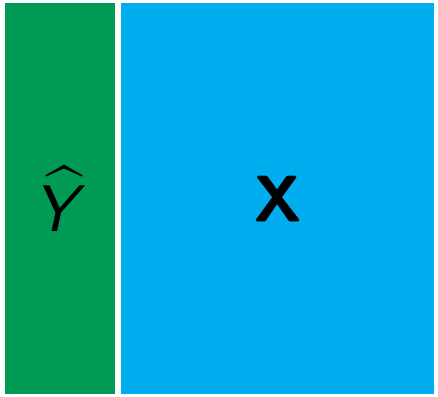

Y: Outcome
$\widehat{Y}$: Prediction
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<u>Key Assumption</u>

Labeled and unlabeled data share the same distribution

An aside on the PBI assumptions

Recent work has focused on weakening the assumptions of:

1. A pre-trained ML model
2. Missingness only in Y
3. Same distribution of labeled and unlabeled data
4. Beyong regression parameters

Analysis option 1: Classical inference

Fit a regression with labeled data

Labeled data

Y: Outcome
$\mathbf{X}$: Covariates

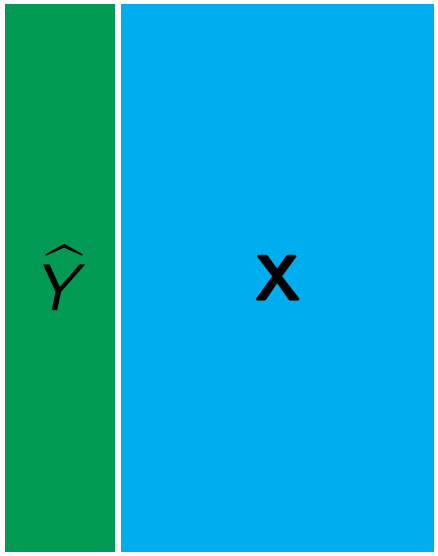
Low bias

High variance

Analysis option 2: Naive inference

Fit a regression with unlabeled data

Unlabeled data



$\widehat{Y}$: Prediction
$\mathbf{X}$: Covariates

High bias

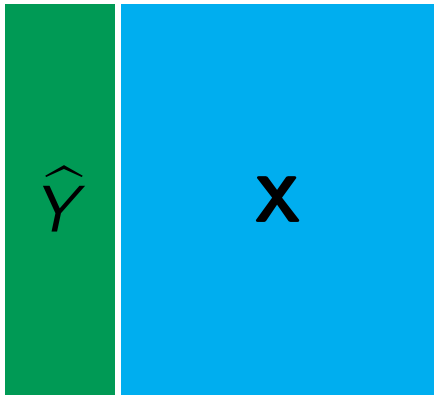
Low variance

Analysis option 3: Prediction-based inference

Use prediction-based methods with all of the data

Labeled data

Unlabeled data



Y: Outcome
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Low bias

Low variance

Back to `postpi` and `PPI`

Question

Do `postpi` and `PPI` mitigate bias from errors in the predictions in regression?

Back to `postpi` and `PPI`

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Do `postpi` and `PPI` mitigate bias from errors in the predictions in regression?

Answer: Post-prediction (`postpi`) inference method

Mitigates bias only under strong assumptions of the ML model

Back to postpi and PPI

Question

Do postpi and PPI mitigate bias from errors in the predictions in regression?

Answer: Post-prediction (postpi) inference method

Mitigates bias only under strong assumptions of the ML model

Answer: Prediction-powered inference (PPI) method

Mitigates bias and provides valid inference irrespective of the quality of the ML model

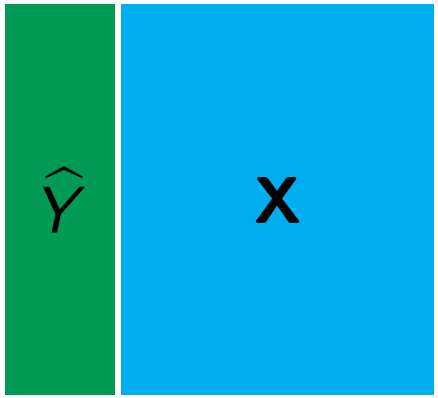
Motwani & Witten. Revisiting inference after prediction. JMLR 2023.

But what about variance?

Use prediction-based methods with all of the data

Labeled data

Unlabeled data



Y: Outcome
$\widehat{Y}$: Prediction
$\mathbf{X}$: Covariates

Low bias

Low variance (?)

Problem set-up

<u>Parameter of interest</u>

$\boldsymbol{\beta}^*$: The population-level regression coefficients of Y on $\mathbf{X}$

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<u>Parameter of interest</u>

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Example: Linear regression

$$\boldsymbol{\beta}^* := \arg\min_{\boldsymbol{\beta}} \mathbb{E}[(Y - \mathbf{X}^\top \boldsymbol{\beta})^2] = [\mathbb{E}(\mathbf{X}\mathbf{X}^\top)]^{-1}\mathbb{E}(\mathbf{X}Y)$$

Why it is not a good idea to do naive inference

$\boldsymbol{\beta}^*$: The population-level regression coefficients of Y on $\mathbf{X}$

$\boldsymbol{\gamma}^*$: The population-level regression coefficients of $\widehat{Y}$ on $\mathbf{X}$

$\boldsymbol{\beta}^*$ is generally different than the target of naive inference $\boldsymbol{\gamma}^*$

Why it is not a good idea to do naive inference

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$$\boldsymbol{\gamma}^* := \arg\min_{\boldsymbol{\gamma}} \mathbb{E}[(\widehat{Y} - \mathbf{X}^\top \boldsymbol{\gamma})^2] = [\mathbb{E}(\mathbf{X}\mathbf{X}^\top)]^{-1}\mathbb{E}(\mathbf{X}\widehat{Y})$$

Back to the problem set-up

<u>Parameter of interest</u>

$\boldsymbol{\beta}^*$: The population-level regression coefficients of Y on $\mathbf{X}$

<u>Classical estimator</u>

$\hat{\boldsymbol{\beta}}^{\text{lab}}$: Estimator of $\boldsymbol{\beta}^*$ using only the labeled data

Example: Linear regression
$\hat{\boldsymbol{\beta}}^{\text{lab}}$ is the familiar OLS estimator

Back to the problem set-up

<u>Parameter of interest</u>

$\boldsymbol{\beta}^*$: The population-level regression coefficients of Y on $\mathbf{X}$

<u>Classical estimator</u>

$\hat{\boldsymbol{\beta}}^{\text{lab}}$: Estimator of $\boldsymbol{\beta}^*$ using only the labeled data

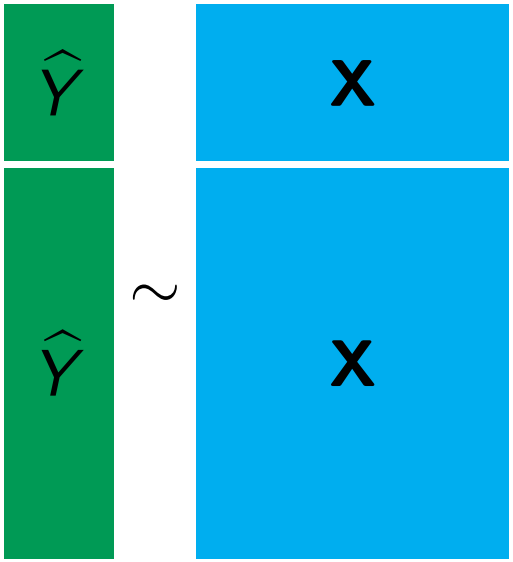
Statistical question

Is the `PPI` method at least as efficient as $\hat{\boldsymbol{\beta}}^{\text{lab}}$?

The PPI estimator for linear regression

Step 1: Obtain the naive estimator

Regress $\widehat{Y}$ on $\mathbf{X}$ with **all of the data** to obtain $\hat{\gamma}^{\text{all}}$



The PPI estimator for linear regression

Step 1: Obtain the naive estimator

Regress $\widehat{Y}$ on $\mathbf{X}$ with **all of the data** to obtain $\hat{\gamma}^{\text{all}}$

Step 2: Obtain the rectifier for bias correction

Use the **labeled data** to compute $\hat{\gamma}^{\text{lab}} - \hat{\beta}^{\text{lab}}$ where

- $\hat{\gamma}^{\text{lab}}$ is from the regression of $\widehat{Y}$ on $\mathbf{X}$
- $\hat{\beta}^{\text{lab}}$ is from the regression of Y on $\mathbf{X}$

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- $\hat{\beta}^{\text{lab}}$ is from the regression of Y on $\mathbf{X}$

Step 3: Correct the naive estimator with the rectifier

Obtain the PPI estimator as $\hat{\beta}^{\text{PPI}} = \hat{\gamma}^{\text{all}} - (\hat{\gamma}^{\text{lab}} - \hat{\beta}^{\text{lab}})$

Intuition for the PPI estimator

PPI estimator

$$\hat{\boldsymbol{\beta}}^{\text{PPI}} = \overbrace{\hat{\boldsymbol{\gamma}}^{\text{all}}}^{\text{Naive estimator}} - \underbrace{(\hat{\boldsymbol{\gamma}}^{\text{lab}} - \hat{\boldsymbol{\beta}}^{\text{lab}})}_{\text{Rectifier}}$$

$\downarrow$ variance with the naive estimator, $\downarrow$ bias with the rectifier

Punchline: PPI can perform worse than classical inference

$\hat{\beta}^{\text{PPI}}$ is not always as efficient as $\hat{\beta}^{\text{lab}}$

Let $\epsilon_Y = Y - \mathbf{X}^\top \beta^*$ and $\epsilon_{\widehat{Y}} = \widehat{Y} - \mathbf{X}^\top \gamma^*$.

For linear regression, $\text{aVar}(\hat{\beta}^{\text{PPI}}) \prec \text{aVar}(\hat{\beta}^{\text{lab}})$ provided that

$$\mathbb{E}\left[\left(\epsilon_{\widehat{Y}} - \epsilon_Y\right)^2 \mid \mathbf{X}\right] < \mathbb{E}\left(\epsilon_Y^2 \mid \mathbf{X}\right).$$

$\widehat{Y}$ must have "enough" predictive power for Y beyond what is explained by the regression of Y on $\mathbf{X}$

Gronsbell, Gao, McCaw, Shi, and Cheng. Another look at statistical inference with ML-imputed data. 2024.

A quick fix: The Chen & Chen (CC) estimator

$$\boxed{\hat{\boldsymbol{\beta}}^{\text{PPI}} = \overbrace{\hat{\boldsymbol{\gamma}}^{\text{all}}}^{\text{Naive estimator}} - \underbrace{(\hat{\boldsymbol{\gamma}}^{\text{lab}} - \hat{\boldsymbol{\beta}}^{\text{lab}})}_{\text{Rectifier}}}$$

$\downarrow$ variance with naive estimator, $\downarrow$ bias with rectifier

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Step 1: Reframe the PPI estimator

$$\hat{\boldsymbol{\beta}}^{\text{PPI}} = \overbrace{\hat{\boldsymbol{\beta}}^{\text{lab}}}^{\text{Classical estimator}} - \underbrace{\left(\hat{\boldsymbol{\gamma}}^{\text{lab}} - \hat{\boldsymbol{\gamma}}^{\text{all}}\right)}_{\text{Augmentation} \xrightarrow{p} 0}$$

A quick fix: The Chen & Chen (CC) estimator

$$\boxed{\hat{\boldsymbol{\beta}}^{\text{PPI}} = \overbrace{\hat{\boldsymbol{\gamma}}^{\text{all}}}^{\text{Naive estimator}} - \underbrace{(\hat{\boldsymbol{\gamma}}^{\text{lab}} - \hat{\boldsymbol{\beta}}^{\text{lab}})}_{\text{Rectifier}}}$$

Step 1: Reframe the PPI estimator

$$\hat{\boldsymbol{\beta}}^{\text{PPI}} = \overbrace{\hat{\boldsymbol{\beta}}^{\text{lab}}}^{\text{Classical estimator}} - \underbrace{(\hat{\boldsymbol{\gamma}}^{\text{lab}} - \hat{\boldsymbol{\gamma}}^{\text{all}})}_{\text{Augmentation} \xrightarrow{p} 0}$$

Step 2: Add a data-adaptive weight to determine how much to borrow from predictions

$$\hat{\boldsymbol{\beta}}^{\text{CC}} = \overbrace{\hat{\boldsymbol{\beta}}^{\text{lab}}}^{\text{Classical estimator}} - \underbrace{\widehat{\mathbf{W}}^{\text{CC}}(\hat{\boldsymbol{\gamma}}^{\text{lab}} - \hat{\boldsymbol{\gamma}}^{\text{all}})}_{\text{Weighted Augmentation}}$$

A quick fix: The Chen & Chen (CC) estimator

$$\hat{\boldsymbol{\beta}}^{\mathsf{CC}} = \hat{\boldsymbol{\beta}}^{\mathsf{lab}} - \widehat{\mathbf{W}}^{\mathsf{CC}}(\hat{\boldsymbol{\gamma}}^{\mathsf{lab}} - \hat{\boldsymbol{\gamma}}^{\mathsf{all}})$$

where $\widehat{\mathbf{W}}^{\mathsf{CC}}$ is an estimator of

$$\mathbf{W}^{\mathsf{CC}} = \mathbf{Cov}(\hat{\boldsymbol{\beta}}^{\mathsf{lab}}, \hat{\boldsymbol{\gamma}}^{\mathsf{lab}} - \hat{\boldsymbol{\gamma}}^{\mathsf{all}}) \left[\mathbf{Var}(\hat{\boldsymbol{\gamma}}^{\mathsf{lab}} - \hat{\boldsymbol{\gamma}}^{\mathsf{all}})\right]^{-1}$$

The CC estimator is optimal in the class of estimators to which PPI belongs

Alternative approach: PPI extensions

Augment the estimating equation rather than the estimator

Example: Linear regression

The familiar OLS estimator solves:

$$n^{-1} \sum_{i=1}^{n} \mathbf{X}_i (Y_i - \boldsymbol{\beta}^\top \mathbf{X}_i) = 0.$$

Several extensions to PPI solve:

$$n^{-1} \sum_{i=1}^{n} \mathbf{X}_i (Y_i - \boldsymbol{\beta}^\top \mathbf{X}_i) - \mathbf{W} \left[N^{-1} \sum_{i=1}^{N} \mathbf{X}_i (\widehat{Y}_i - \boldsymbol{\beta}^\top \mathbf{X}_i) - n^{-1} \sum_{i=1}^{n} \mathbf{X}_i (\widehat{Y}_i - \boldsymbol{\beta}^\top \mathbf{X}_i) \right] = 0$$

where $\mathbf{W}$ is a data-adaptive weight matrix.

Alternative approach: PPI extensions

- `PPI++`: Selects **W** to be a scalar to minimize MSE
- `PSPA`: Selects **W** to be a diagonal matrix to minimize element-wise variance
- `PDC`: **Selects W to be a full matrix to minimize element-wise variance**

Practical Recommendations

Method	Low bias?	Lower variance?	Full matrix?
`PostPI`	?	✓	✗
`PPI`	✓	?	Identity
`PPI++`	✓	✓	Scalar
`PSPA`	✓	✓	Diagonal
`PDC`	✓	✓*	✓
`CC`	✓	✓†	✓

$*$ optimal within the PPI extension class

$\dagger$ optimal within the linear augmentation class

Goals for this tutorial

1. Understand the fundamentals of PBI
2. Gain familiarity with PBI methods
3. Learn to run an analysis with predicted data using `ipd`

   → Let's get started! Please go to: https://salernos.github.io/ipd-workshop/.

Supplemental Simulated example

Generative model

$$Y = X_1 + X_2 + 0.5X_3 + 0.5X_4 + 0.25X_5 + \epsilon$$

Scientific goal

Evaluate the association between Y and X_1

Set-up

Compare the performance of 3 methods used to evaluate the association with a dataset of size 10,000 with varying proportions of labeled data (10-40%)

Simulated example

Methods

- Naive: Regression of $\widehat{Y}$ on X_1 with all of the data
- Classical: OLS regression of Y on X_1 with labeled data
- `PPI`: Prediction-based regression with all of the data

Prediction model settings

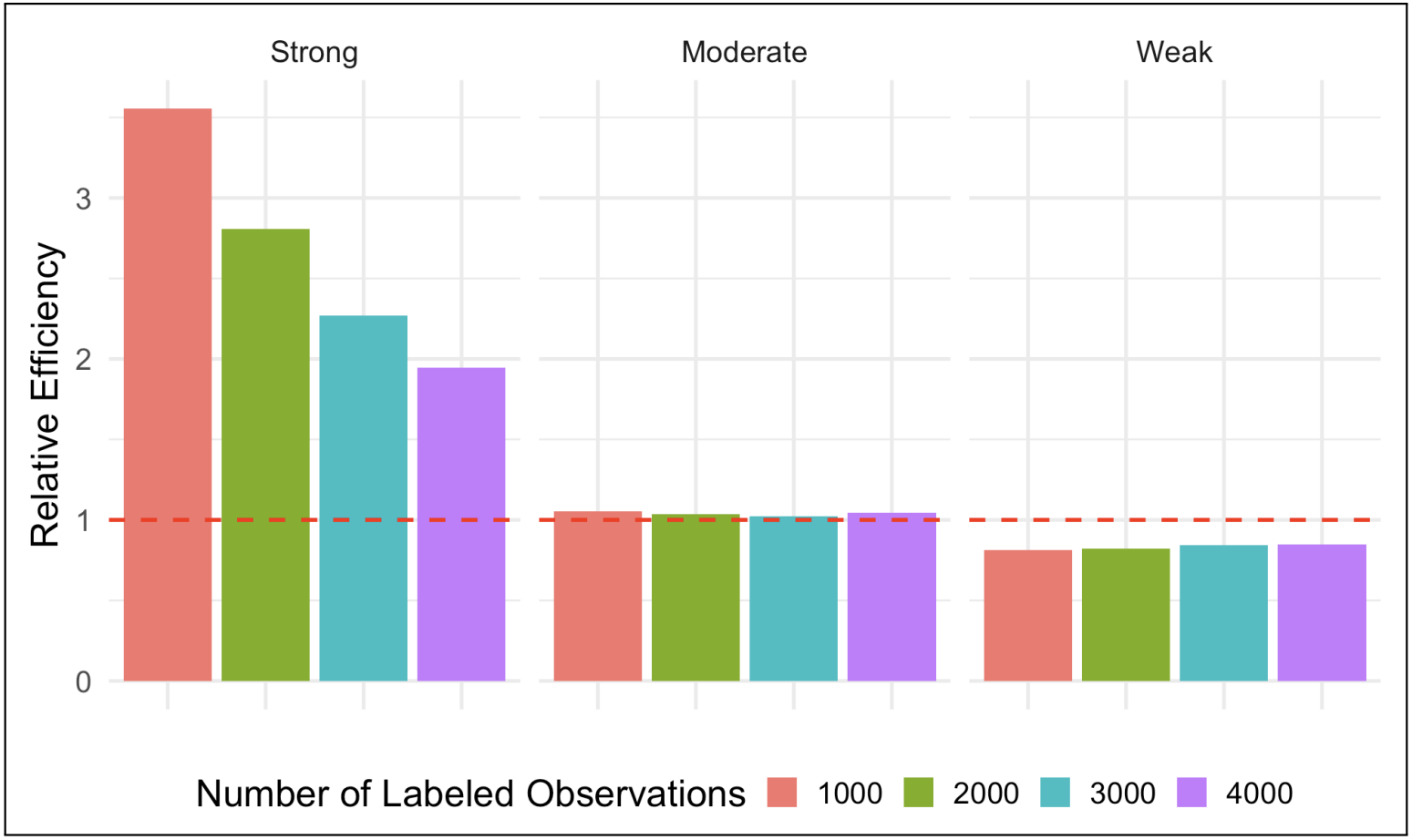
Strong $\widehat{Y}$	Moderate $\widehat{Y}$	Weak $\widehat{Y}$
$R^2_{\widehat{Y},Y} = 0.9$	$R^2_{\widehat{Y},Y} = 0.7$	$R^2_{\widehat{Y},Y} = 0.5$

Simulated example: Percent bias

	Labeled set size			
Method	1000	2000	3000	4000
Strong $\widehat{Y}$				
Naive	-1.205	-1.205	-1.205	-1.205
Classical	0.008	0.002	-0.016	-0.001
PPI	0.007	0.000	-0.007	-0.006
Moderate $\widehat{Y}$				
Naive	-46.980	-46.980	-46.980	-46.980
Classical	-0.033	-0.023	0.027	0.004
PPI	-0.024	-0.012	0.006	-0.018
Weak $\widehat{Y}$				
Naive	-64.272	-64.272	-64.272	-64.272
Classical	-0.012	-0.003	0.002	0.012
PPI	0.000	0.001	-0.006	0.012

Naive inference can result in severe bias

Simulated example: Relative efficiency



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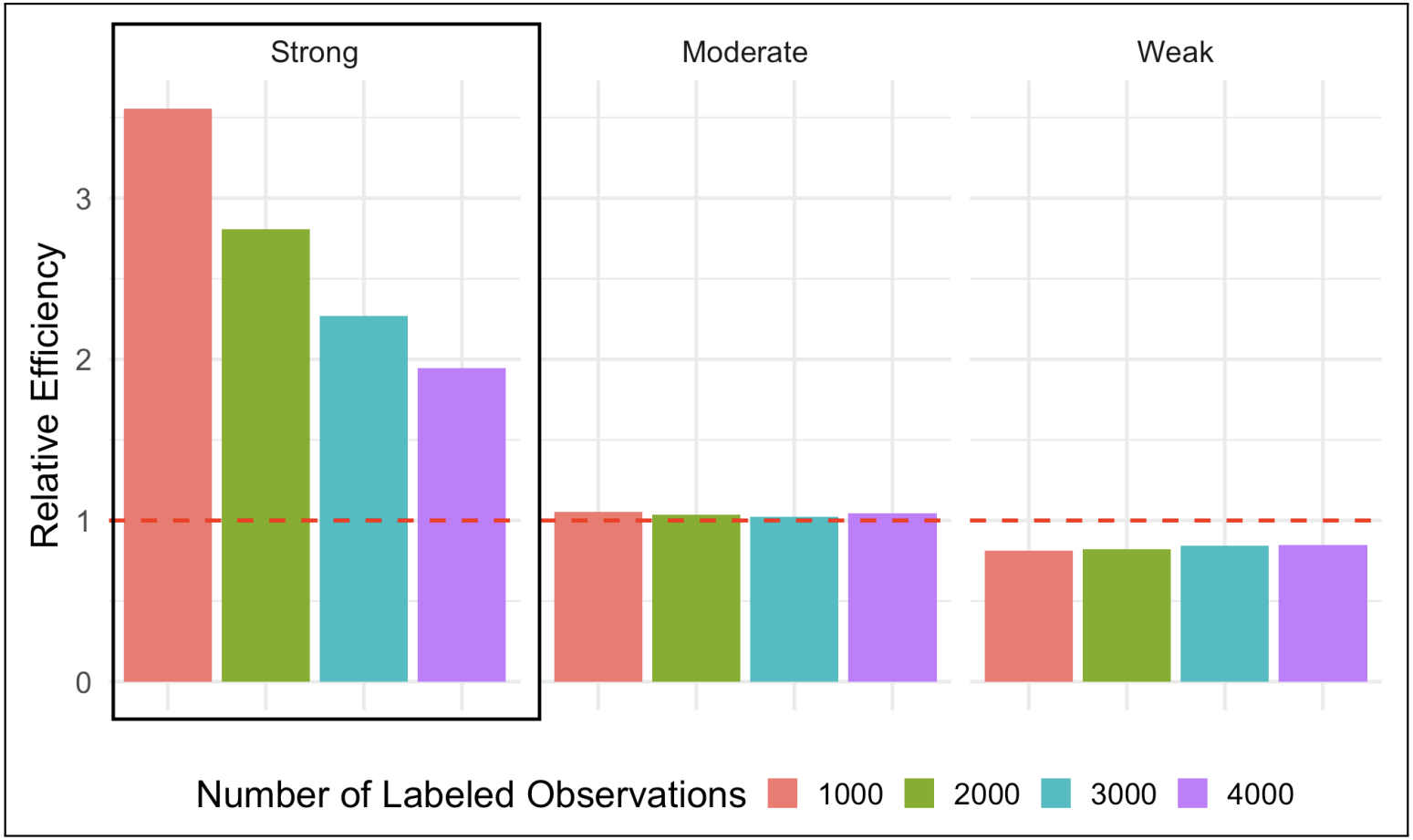
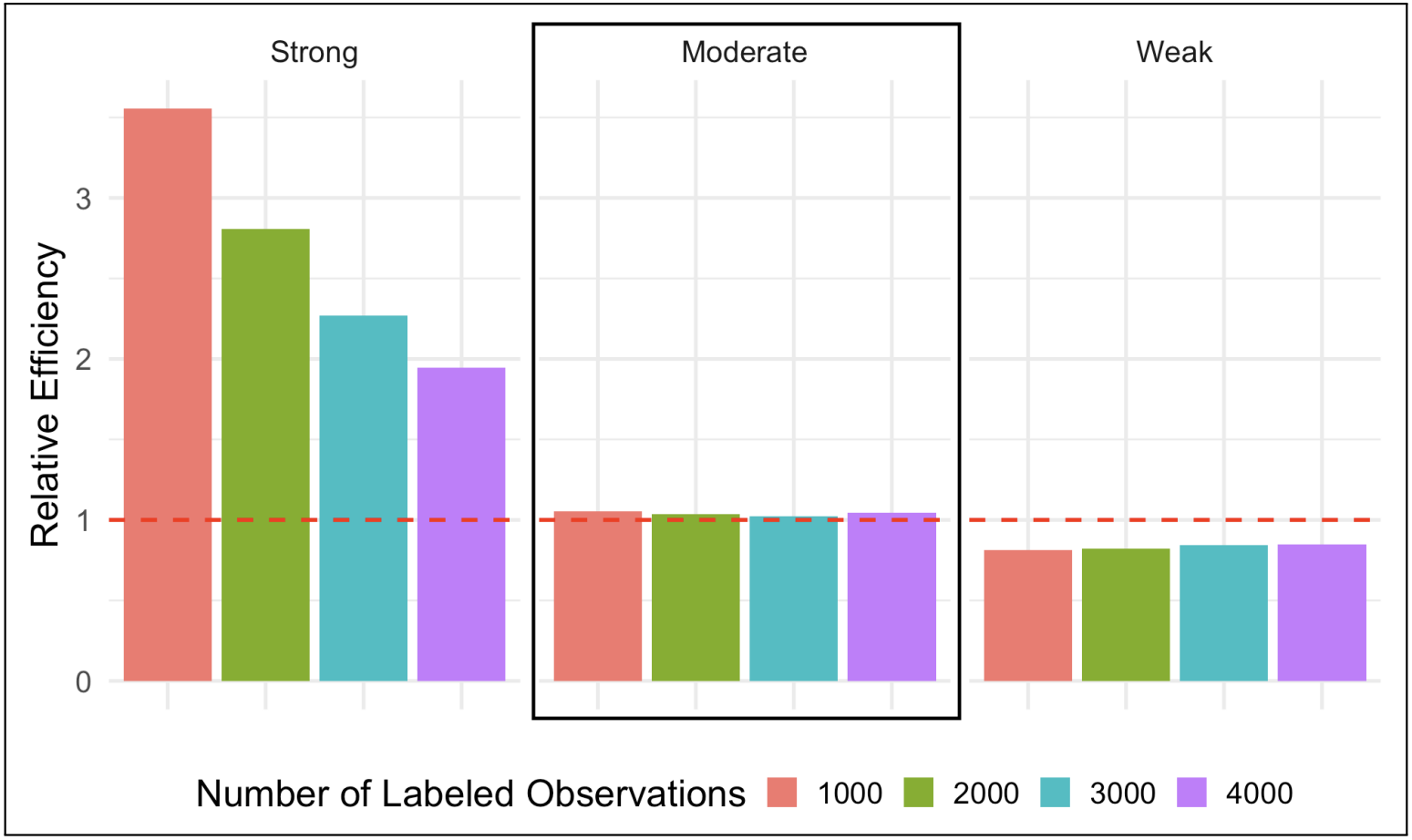


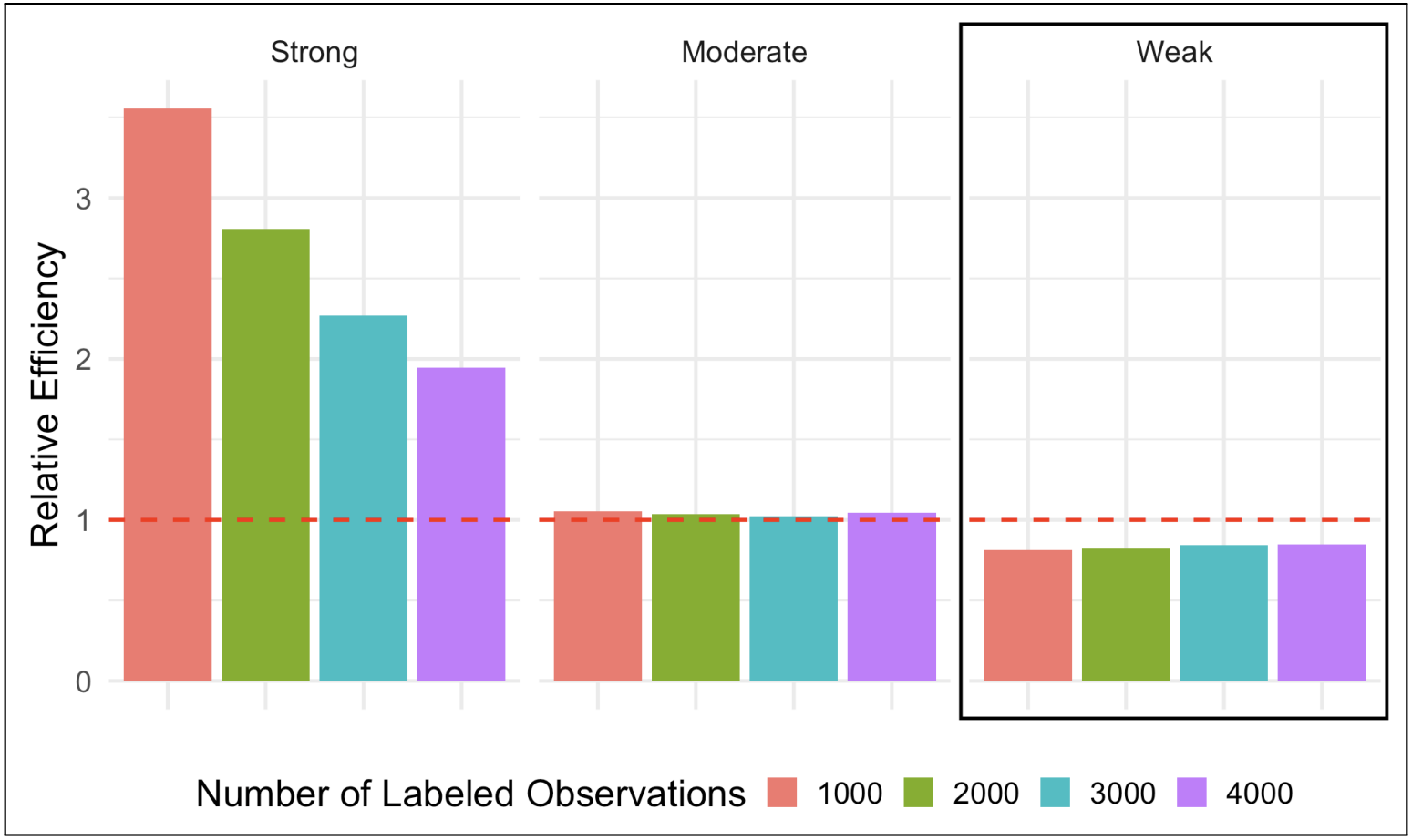
With a strong prediction, variance is 2-3x lower with `PPI`

Simulated example: Relative efficiency



With a moderate prediction, you can just do classical inference

Simulated example: Relative efficiency



With a weak prediction, `PPI` discards 20% of the labeled data

Back to the simulated example: Percent bias

	Labeled set size			
Method	1000	2000	3000	4000
Strong $\widehat{Y}$				
Naive	-1.205	-1.205	-1.205	-1.205
Classical	0.008	0.002	-0.016	-0.001
PPI	0.007	0.000	-0.007	-0.006
CC	0.006	0.000	-0.007	-0.006
Moderate $\widehat{Y}$				
Naive	-46.980	-46.980	-46.980	-46.980
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PPI	-0.024	-0.012	0.006	-0.018
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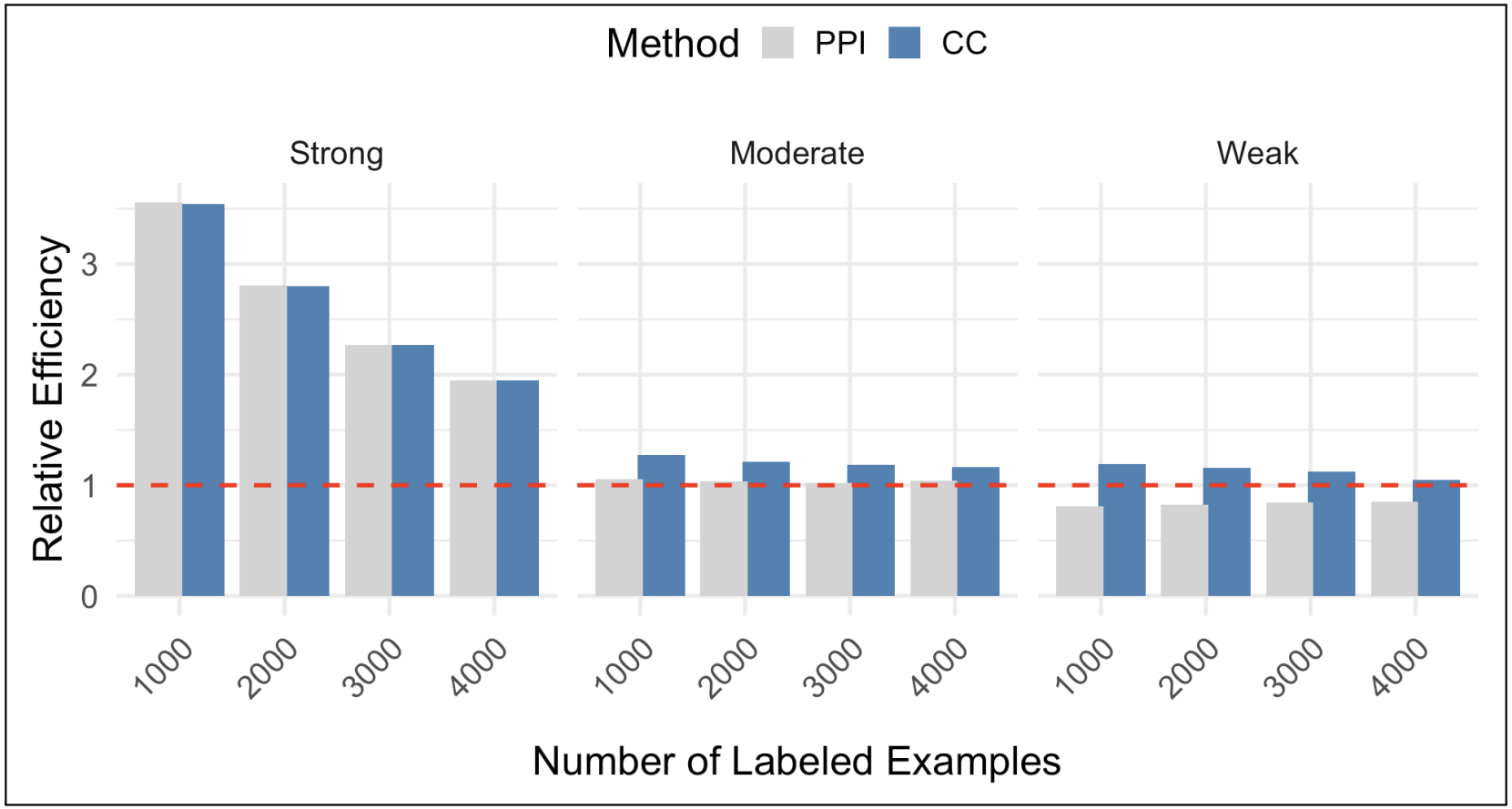
Weak $\widehat{Y}$				
Naive	-64.272	-64.272	-64.272	-64.272
Classical	-0.012	-0.003	0.002	0.012
PPI	0.000	0.001	-0.006	0.012
CC	-0.014	-0.006	-0.003	0.011

The CC estimator yields unbiased estimation

Back to the simulated example: Relative efficiency



CC inference has higher precision than classical inference